Chapter 9

VLAN, VTP

Virtual LAN (VLAN) is technique of logically grouping of computers of a LAN to limit broadcast domain and improve security. The first thing that might come to mind when we mention VLANs is that somehow they don't exist physically. They do exist and they are very commonly used in the enterprise or corporate computing networks to segment networks. In the past, to separate or segment networks, we simply used separate pieces of hardware such as hubs or switches. It became very clear that when hubs and switches only come in predefined capacities such as 5, 8, 12, 24, or even 36 ports, potentially we could be wasting resources by not fully utilizing all the ports on these devices. In addition, as companies expanded and spread into different locations, it became difficult to have people in the same departments on the same LAN segments. Someone envisioned being able to change and reconfigure one device into multiple segments and VLANs were born.

Broadcast Domains

One of the problems that we must control in a network environment is the spread of broadcasts. A broadcast is a data transmission or packet that is sent to every machine. Each machine must process this packet and see whether it relates to them. An analogy might be: to hear an announcement on the PA, we all must pay attention until the point when we realize it has no importance to us. In a network, broadcasts take up valuable bandwidth that could be used to carry other data and also take up valuable processing power to determine whether the packet is important or not. A broadcast uses a very special Media Access Control (MAC) address of FF:FF:FF:FF:FF:FF. As we saw in Chapter 11, a networking device listens for its own MAC address and for the broadcast MAC address and processes these packets only. Because switches only store source MAC address in their switching tables, a broadcast MAC address is never stored in the switching table. When a switch receives a packet that is not contained in its switching table, it floods that packet to every port to find the recipient. This means that the switches do not impede any broadcasts. If we were going "old school" and using hubs, hubs merely repeat the packet to every single port. Only routers impede and do not forward broadcasts. The area of the network that is defined when a broadcast propagates from one switch or hub to another and one PC to another is known as a "broadcast domain." The more the broadcasts on the network, the less efficient and productive the network is. There are certain networking services that rely on broadcasts that are very common such as Dynamic Host Configuration Protocol (DHCP) or Address Resolution Protocol (ARP) and therefore all broadcasts can't be eliminated. To control broadcasts, we segment networks into different pieces to control how many broadcasts are on each segment. VLANs are a natural extension to segment a network physically by segmenting the network virtually. This allows us maximum flexibility in configuration of segmenting the network to control broadcast domains.

Companies often use VLANs in security situations as well. For a long time, Cisco did not officially endorse using VLANs for security but in recent documentation does talk about this. How can we use VLANs to help security? Because we are virtually segmenting the network into separate segments, we can group different networking devices on segments based on their role in our organization. For example, we might have a VLAN that specifically was set up for resources in the Human Resources (HR) department. We could control access to this particular segment so that only those PCs that require access to the HR servers can get access to them. Most organizations do use VLANs to some extent to segment sensitive resources from other VLANs on the network.

9.1 VLAN Configuration

Configuring a VLAN in a Cisco switch is not an overly difficult process. We'll walk through the basic steps of the process, which include setting up the different numbered or named VLANs and setting the IP address for each VLAN.

Naming and Numbering the VLAN

Cisco VLANs are numbered from 1 to 4,094. The normal VLANs are numbered from 1 to 1,005. These VLANs are stored in the switches flash memory in a file called vlan.dat. In addition, there are VLANs numbered from 1,006 to 4,094 that are called the extended VLANs. The extended VLANs do not have as many options and are stored in the running configuration file of the switch. There are several VLANs that are set up in Cisco switches by default. These are set at the factory and cannot be changed by the end user. These include what we call the administrative VLAN, which is VLAN 1, and VLANs 1,002–1,005, which are for Token Rin and FDDI networks, respectively. To enable easier identification, we may also name VLANs so that they have a better description of their use.

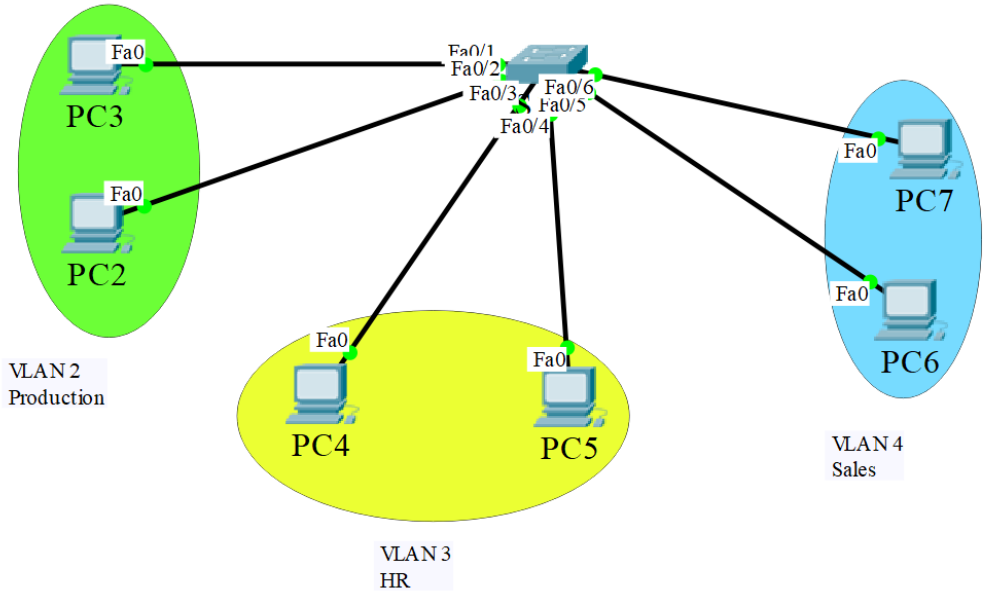


Fig. 9.1 Physical topology of a network

Let's consider an office which consists of three departments Production, HR and Sales. We want to limit traffic of each department within each department. To enable this feature, we have to create VLAN for each department as shown in Fig. 9.1. Next, we will see how we can create VLAN. To access the VLAN configuration menu, there are a series of steps we need to perform. We must initially enter the Global Configuration mode by typing configure terminal command.

1. First create a VLAN. To create a VLAN, we have to use the following command:
   ```
   Switch(config)#vlan 2
   ```

   where 2 is the VLAN number. Every VLAN must have a VLAN number. The number can range 1 to 1005. It is optional to give a name to a VLAN. After creating a VLAN, we can name it. To name the VLAN 2 as Production, we can use the following command.

   ```
   Switch(config-vlan)#name Production
   ```

   If we want to create multiple VLANs and name them, we have to use the above two commands for each VLAN. Thus, to create a VLAN for HR department, we can write

   ```
   Switch(config)#vlan 3
   Switch(config-vlan)#name HR
   ```

   Pertinently, it is not required to number VLAN sequentially. We can choose any value in the range mentioned above to number a VLAN.

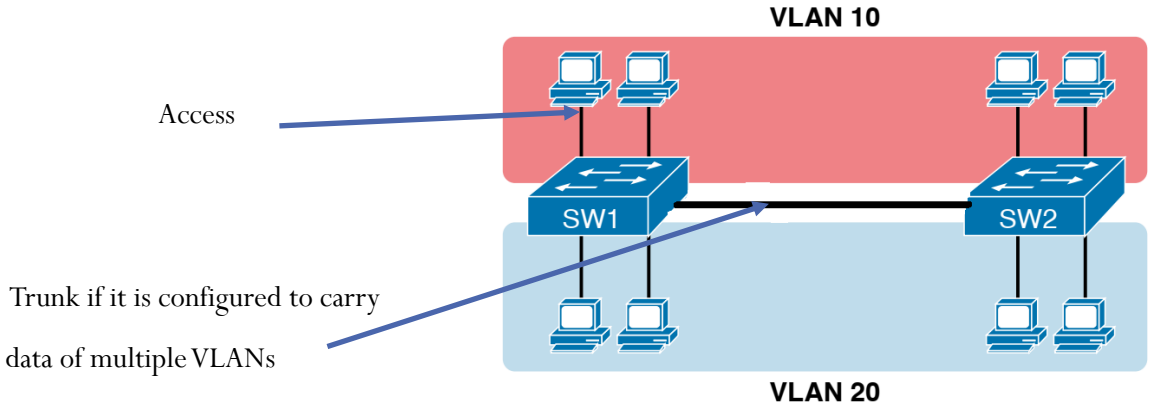


Fig.9.2 Access and Trunk modes of switchports

2. After creating a VLAN, we have to specify which interfaces (also known as ports) of a switch will belong to the VLAN. To assign interfaces to a VLAN, we can add one interface at a time or multiple interfaces at a time. To add one interface at a time, select an interface. Suppose that we want to add fa0/1 interface to the VLAN 2. We can do this by the following commands:

   ```
   Switch(config)#interface fa0/1
   ```

```
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
```

The first line selects the interface fa0/1. The second line specifies the purpose for which the fa0/1 interface of the VLAN 2 will be used. An interface of a switch can be used to carry data of a single VLAN or multiple VLANs (see Fig. 9.2). If we decide to use an interface for former case (for example, connecting a PC to a VLAN), the mode of the switchport must be `access`. For the latter one, it should be `trunk`. Since we are going to connect PC3 to the fa0/1 interface, we have made the mode of the interface access in the second line. In the third line, we add the fa0/1 interface to the VLAN 2. To add the fa0/2 interface same VLAN, we have to repeat the same procedure.

```
Switch(config-if)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
```

However, there is a shortcut way of adding multiple interfaces to a VLAN by changing the first line a bit. To add fa0/3 and fa0/4 to VLAN 3, we have to use the following command:

```
Switch(config-if)#int range fa0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 3
```

Similarly, we can add fa0/5 and fa0/6 interfaces to the VLAN4.

We can verify whether the VLANs have been created successfully and the assignment of the interfaces to different VLAN has been done as intended. This can be done using `show vlan brief` command in privileged mode. As we see in Fig. 9.3, Fa01/ and Fa0/2 interfaces have been included in the VLAN having number 2 and name Production. We also see that the interfaces except Fa0/1-6 still belong to the default VLAN (VLAN 1).

```
Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/1, Gig0/2
2    Production                       active    Fa0/1, Fa0/2
3    HR                               active    Fa0/3, Fa0/4
4    Sales                            active    Fa0/5, Fa0/6
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Fig.9.3 Verification of VLAN

The `show vlan brief` command shows all VLANs. If we want to observe any particular VLAN, we can specify the VLAN using its number or name. Fig. 9.4 shows the VLAN 2.

```
Switch>en
Switch#show vlan id 2

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
2    Production                       active    Fa0/1, Fa0/2

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
2    enet  100002     1500  -      -      -        -    -        0      0

Switch#show vlan name Production

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
2    Production                       active    Fa0/1, Fa0/2

VLAN Type  SAID       MTU   Parent RingNo BridgeNo Stp  BrdgMode Trans1 Trans2
---- ----- ---------- ----- ------ ------ -------- ---- -------- ------ ------
2    enet  100002     1500  -      -      -        -    -        0      0
```

Fig. 4 Showing the properties of a particular VLAN

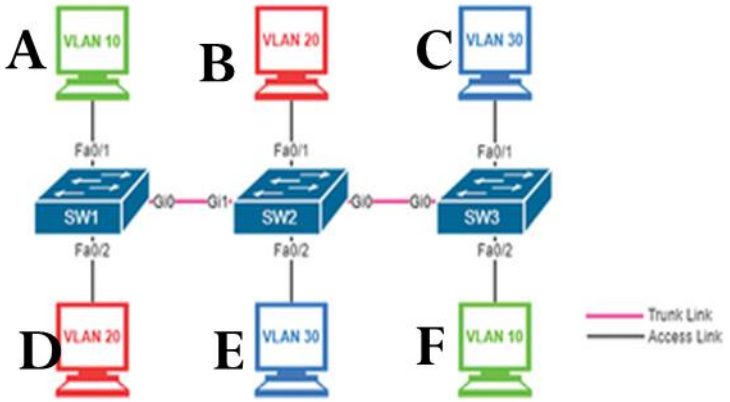


Fig. 9.5 Network for configuring VTP

9.2 VLAN Trunking Protocol (VTP)

Let's consider a network given in Fig. 9.5. In the network, we see that the same VLAN exists in multiple switches. For example, VLAN 20 exists in SW1 and SW2. On the other hand, VLAN 20 exists in SW1 and SW 3. We know that a PC in one VLAN can communicate with another PC in the same VLAN. However, is it possible for a PC in a VLAN in a switch to communicate with

another PC of the same VLAN connected to another switch? Yes, it can. However, we have to connect the two switches using trunk, a link between two switches that carries frames of multiple VLANs. For example, to enable PC D of VLAN 20 connected to SW1 to communicate the PC B of VLAN 20 connected to SW2, the link between SW1 and SW2 must be configured as a trunk. To configure a link as a trunk, interfaces of both sides of the link have to be configured as trunks. Thus, the fa0/0 interface of SW1 and fa0/1 of SW2 must be configured as trunk. To configure the fa0/0 interface of SW1, use the following command in SW1's global configuration mode:

```
SW1(config)#interface fa0/0

SW1(config-if)#switchport mode trunk
```

Similarly, we have to configure the fa0/1 interface of SW2 as

```
SW2(config)#interface fa0/1

SW2(config-if)#switchport mode trunk
```

Following the same procedure, we have to configure the link between SW2 and SW3 as a trunk.

Now, to configure VTP in each switch, we have to set values of three parameters: mode, domain and/or password. VTP has three modes: server, client and transparent. The server mode is a default mode. However, we have to change the mode if we want to make the mode client or transparent. To set the mode, use the following command in global configuration mode:

```
SW1(config)#vtp mode server
```

or

```
SW1(config)#vtp mode client
```

or

```
SW1(config)#vtp mode transparent
```

The domain is any word. The server and all clients must have the same domain. To set the domain to *aiub* (suppose), use the following command:

```
SW1(config)#vtp domain aiub
```

Password is optional. If we configure a password for VTP, we must configure the password on all switches in the VTP domain. The password must be the same password on all those switches. We can set a password as follows:

```
SW1(config)#vtp password 123
```

In this case, we have set the password 123. The password can consist of up to 64 characters.

If we want to configure SW1 as a server and SW2 and SW3 as clients, we can figure them as follows:

SW1

```
Switch(config)# vtp mode server

Switch(config)# vtp domain aiub

Switch(config)# vtp password 123
```

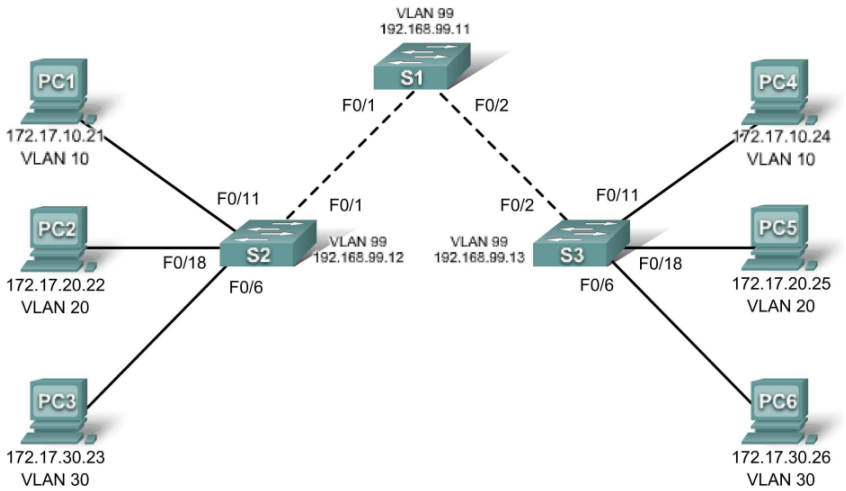
SW2

```
Switch(config)# vtp mode client
Switch(config)# vtp domain aiub
Switch(config)# vtp password 123
```

SW3

```
Switch(config)# vtp mode client
Switch(config)# vtp domain aiub
Switch(config)# vtp password 123
```

If we want to configure VTP in a switch in transparent mode, we have to create VLANs manually in the switch. If we want a switch to pass frames of the VLAN 10, we have to create that VLAN 10 in the switch manually even if the switch does not need to add any of its interfaces to that VLAN. In Fig. 9.5, suppose that we want to configure VTP in SW2 as transparent. To enable communication between PCs A and F which are under the same VLAN but connected to the SW1 and SW3, respectively, SW2 needs to pass the frame between the PCs. In this case, SW2 must have the VLAN 10 even though it does not use the VLAN 10 for itself.

<u>Exercise</u>

Configure VLAN and VTP in the following network, where the SW1 must be configured in transparent mode.